

cell X

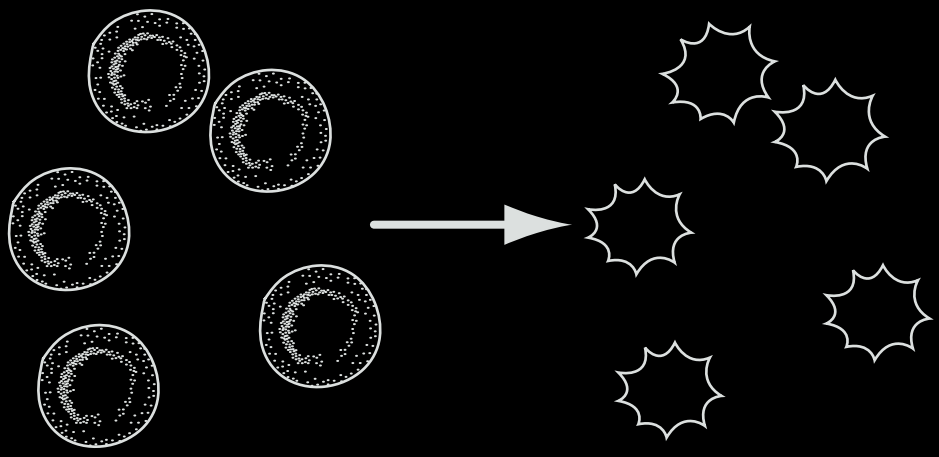


cell Y

Plant cells become turgid when they take up water by osmosis.

In which of these cells does being turgid provide support?

	cell X	cell Y
A	no	no
B	no	yes
C	yes	no
D	yes	yes



fresh blood

blood after mixing with liquid **X**

Which statement describes the water potential of liquid **X**?

A It is equal to that of pure water.

B It is equal to that of the cell cytoplasm.

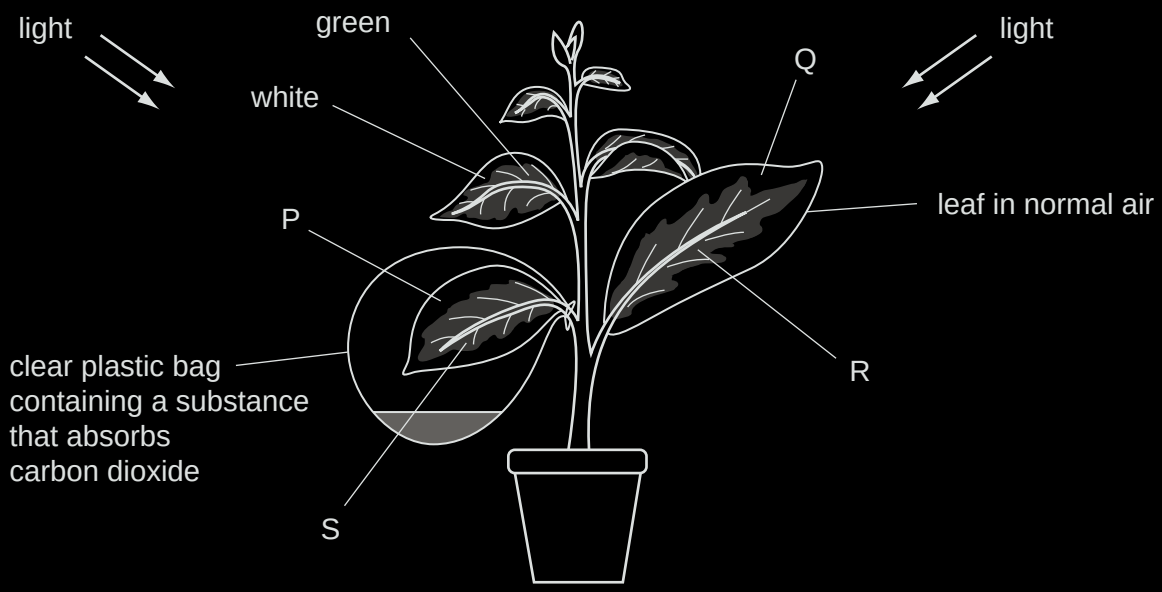
C It is higher than that of the cell cytoplasm.

D It is lower than that of the cell cytoplasm.

4 Starch is digested to maltose by the enzyme amylase.

According to the 'lock and key' hypothesis, which is the 'key' and which is the 'lock'?

	'key'	'lock'
A	amylase	maltose
B	amylase	starch
C	starch	amylase
D	starch	maltose



Which leaf areas lack only **one** factor needed for photosynthesis?

A P and Q

B P and R

C Q and S

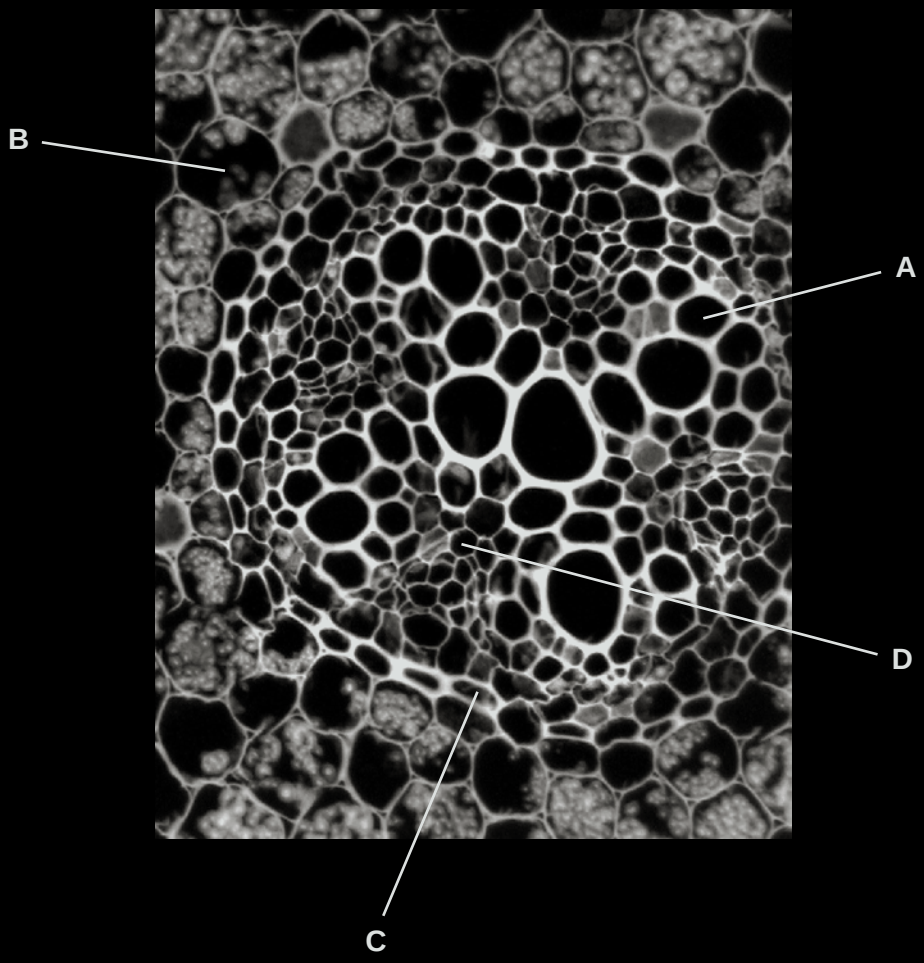
D R and S

6 When is carbon dioxide absorbed and when is it released by an ecosystem, such as a tropical rainforest?

	daylight	darkness
A	absorbed	absorbed
B	absorbed	released
C	released	absorbed
D	released	released

7 What is the function of each type of plant cell?

	palisade cells	phloem cells	root hair cells
A	photosynthesis	sugar transport	ion uptake
B	photosynthesis	sugar transport	transpiration
C	transpiration	photosynthesis	ion uptake
D	transpiration	photosynthesis	sugar transport



12 During translocation in plants, what substance is moved from which organ to which organ?

	substance	from	to
A	sucrose	anthers	stigmas
B	sucrose	leaves	roots
C	water	roots	leaves
D	water	soil	root hairs

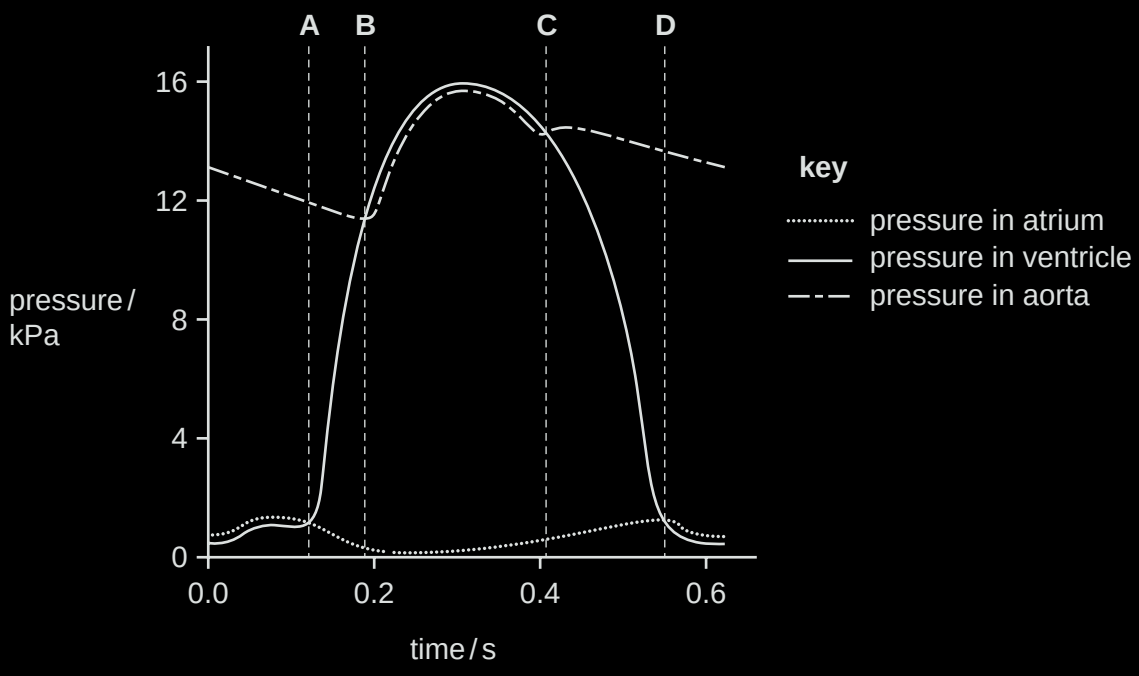
13 Which chambers of the human heart contain oxygenated blood?

A left atrium and left ventricle

B left atrium and right ventricle

C right atrium and left ventricle

D right atrium and right ventricle



15 What causes the transfer of materials between capillaries and tissue fluid?

A active transport

B blood pressure

C capillarity

D osmosis

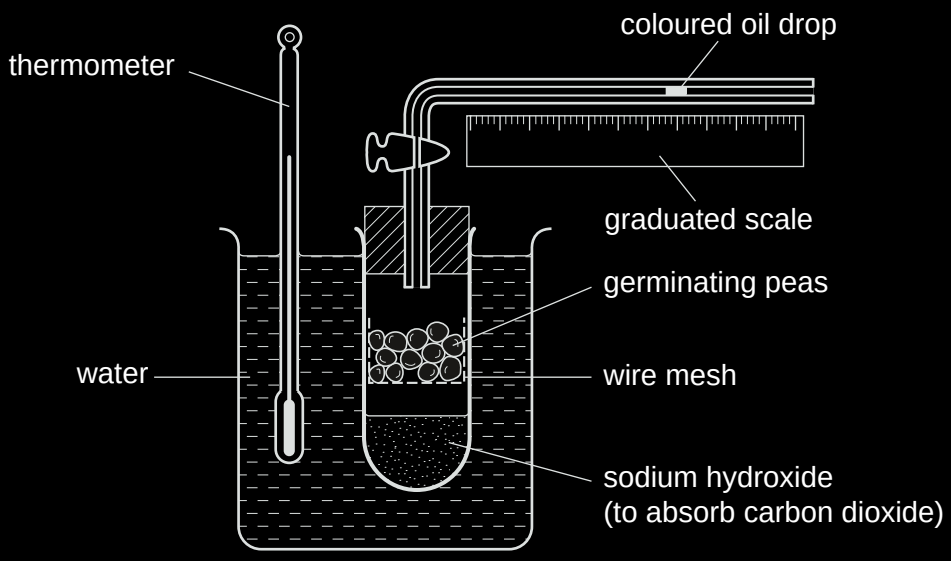
16 In the body of a human, which processes consume energy from respiration?

	absorption of water	cell division	protein synthesis
A	✓	✓	✓
B	✓	✓	✗
C	✓	✗	✓
D	✗	✓	✓

key

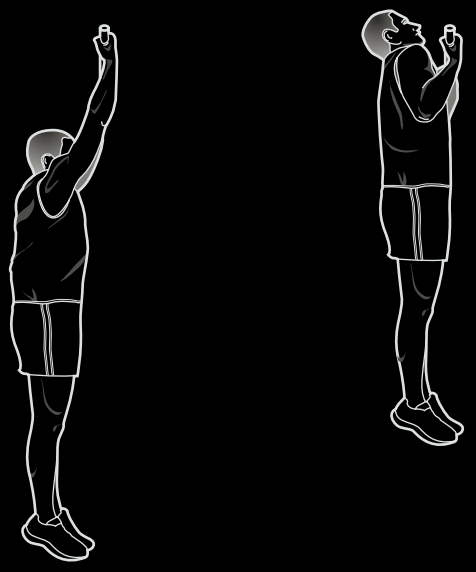
✓ = needs energy

✗ = does not need energy



What change will be seen and what is the explanation?

	change	explanation
A	oil drop moves left	oxygen is used up by the peas
B	oil drop does not move	oxygen is used up as fast as carbon dioxide is released
C	oil drop does not move	carbon dioxide is absorbed
D	oil drop moves right	peas release carbon dioxide



X Y

In moving from the lower position X to the upper position Y, which muscles and joint movements occur?

	biceps	triceps	ball and socket joint
A	contracts	relaxes	rotates
B	contracts	relaxes	extends
C	relaxes	contracts	rotates
D	relaxes	contracts	flexes

24 Which of these substances do **not** occur naturally in the human body?

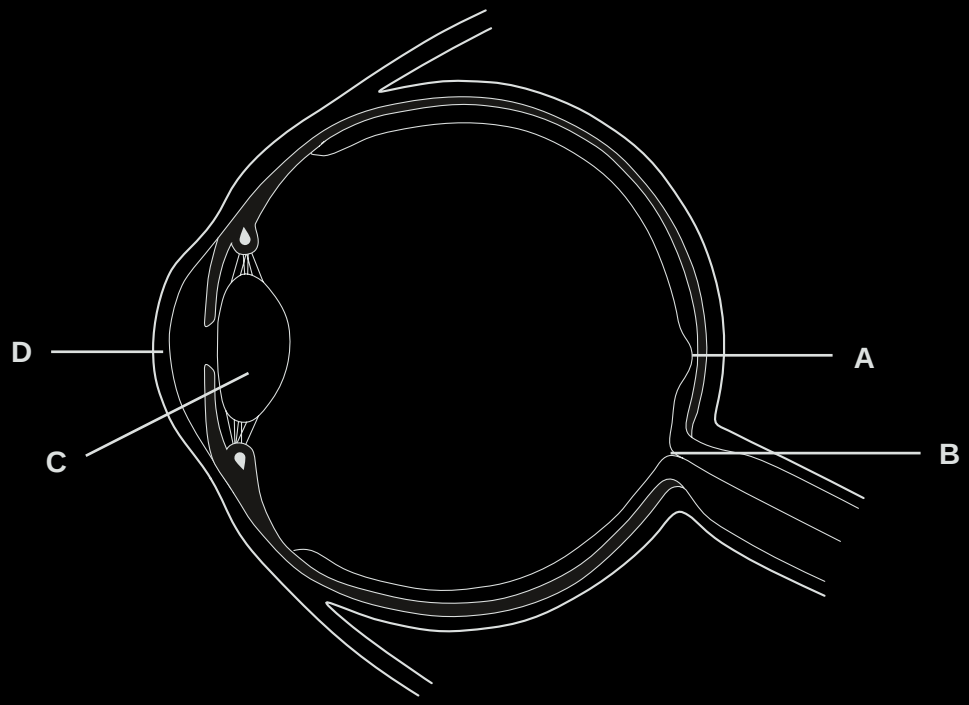
adrenaline	heroin	nicotine	penicillin
1	2	3	4

A 2 and 3 only

B 1, 2 and 3 only

C 1, 3 and 4 only

D 2, 3 and 4 only



26 'This group of organisms have long thread like bodies. Their cells contain nuclei. Some of them produce antibiotics.'

Which group of organisms is being described?

A bacteria

B fungi

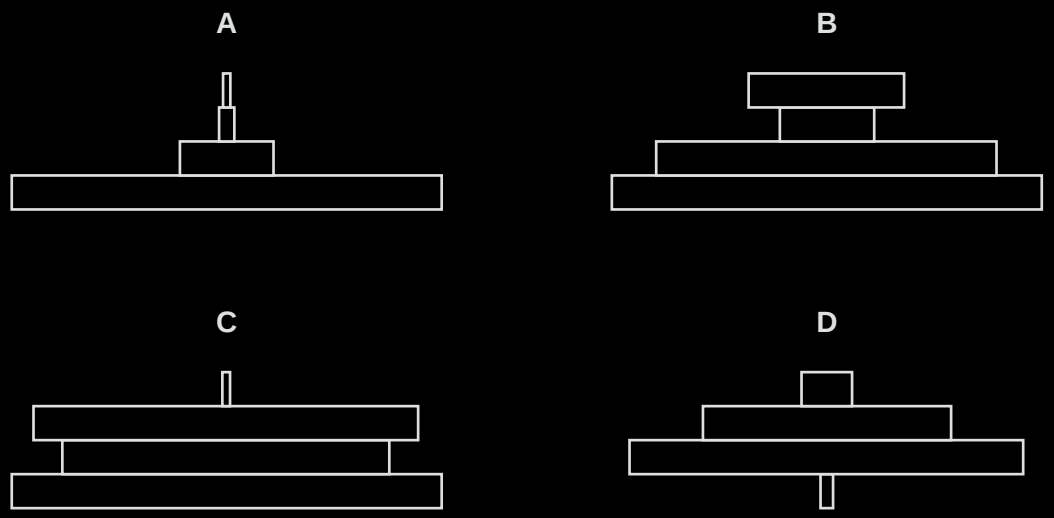
C plants

D viruses

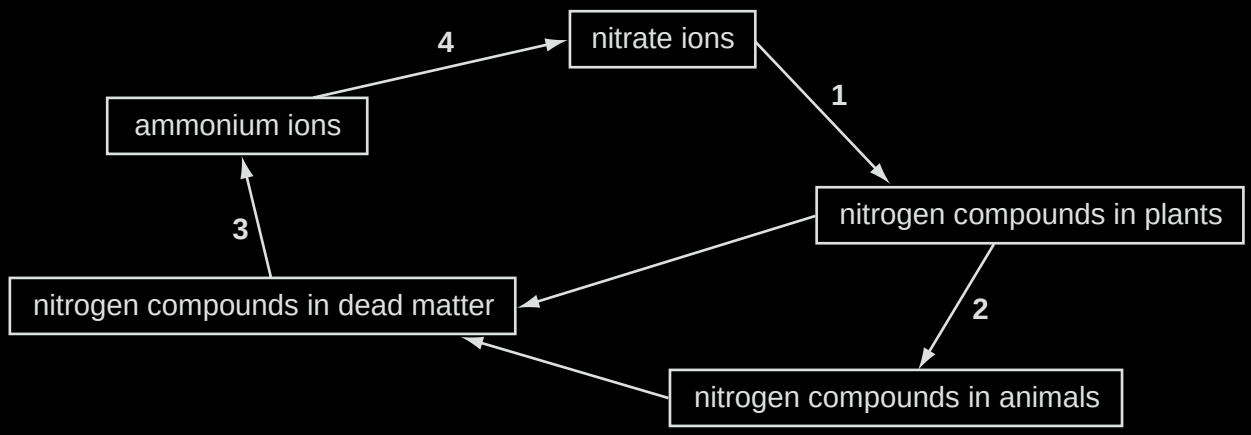
27 Fermenters that are used to produce single cell protein may be closed (batch) or open (continuous).

Which set of fermenter properties is correct?

	closed	open
A	growth rate of culture remains at optimum	fermenting microorganisms must be added all the time
B	growth rate of culture remains at optimum	product can be removed all the time
C	vessel must be sterilised after each separate cycle	fermenting microorganisms must be added all the time
D	vessel must be sterilised after each separate cycle	product can be removed all the time



30 The diagram shows part of the nitrogen cycle.



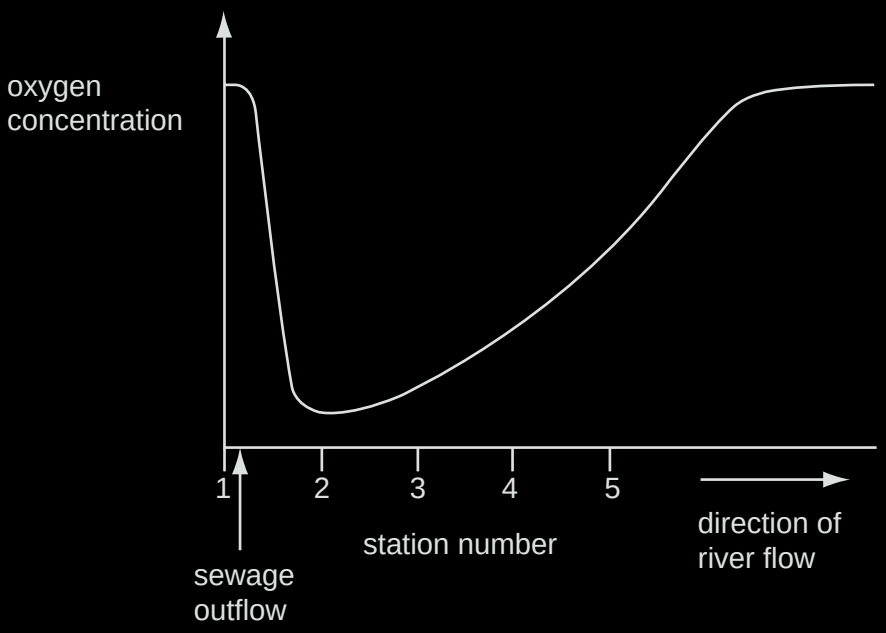
Which stages depend on the action of bacteria?

A 1 and 2

B 2 and 3

C 3 and 4

D 4 and 1



At which stations are the concentrations of organic matter **lowest**?

A 1 and 5

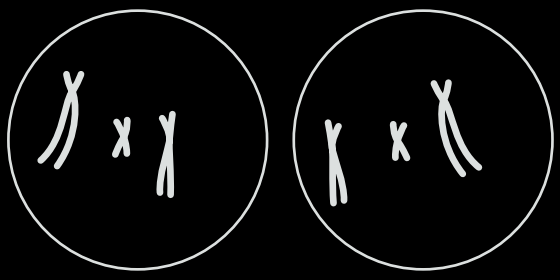
B 2 and 3

C 3 and 4

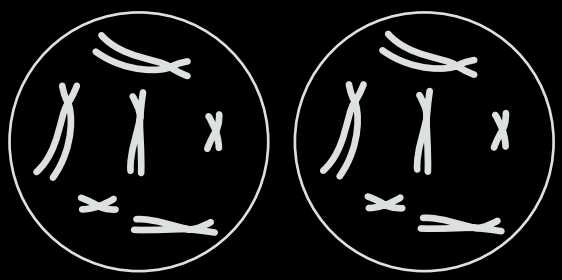
D 4 and 5



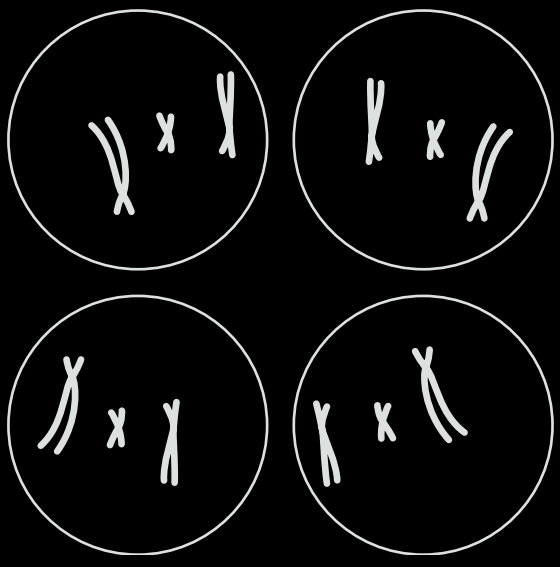
A



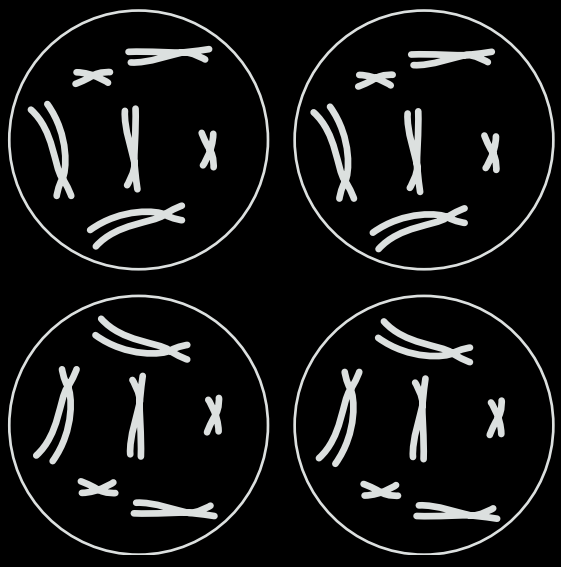
B

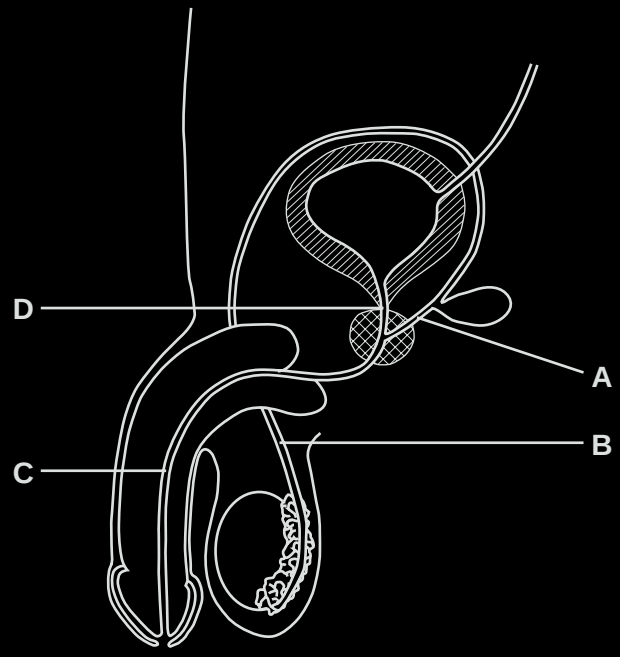


C



D





36 Which diseases can be cured with antibiotics?

	gonorrhoea	HIV infection	syphilis
A	✓	✓	✓
B	✗	✓	✗
C	✓	✗	✓
D	✗	✗	✓

key

✓ = can be cured with antibiotics

✗ = cannot be cured with antibiotics